

Deepak Mallampati

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Kent, OH (Open to SF relocation; F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Forward Deployed AI engineer shipping agentic workflow systems into regulated enterprise (healthcare today, finance-adjacent by pattern). Production RAG with ~35% retrieval precision gains and >90% grounded alignment, multi-agent pipelines with schema-validated JSON contracts that cut hallucinations >30%, and FastAPI + Docker packaging that reduced post-deploy defects ~30%. HIPAA-conscious data governance, HITL escalation, and audit-ready reasoning traces map cleanly to banking compliance vocabulary. Master's in Computer Science, Kent State. Ready to deploy and train AI agents at financial institution clients.

CORE COMPETENCIES

Agentic Workflows • RAG + Grounding • Schema Contracts • HITL Escalation • Audit-Ready Reasoning • Client-Facing Delivery • Python / FastAPI • Docker / CI-CD • PostgreSQL / SQL • Regulated-Domain AI

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agentic Workflows, RAG, Production LLM

USA - Healthcare Technology, AI Consulting (HIPAA-conscious)

- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; response stability ~25%, hallucinations cut >30%.
- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision ~35%, reduced irrelevant retrieval >30%, response alignment >90%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall +15-20% on high-risk categories, precision held >90%.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~30%.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, system-limitation docs for non-technical stakeholders.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP (compliance-sensitive)

- Optimized **T-SQL stored procedures** on the Synthesis Order-to-Cash platform (contracts, allocations, invoicing); query time -20%, retrieval latency -25%.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors -30%, release cycle -35 to -40%.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence -25%, root-cause resolution -30%.
- Supported **RBAC and audit logging** for financial modules in a regulated environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ sessions; manual review cut 60-70%, AI highlights aligned with human curation ~85%.
- Implemented **document Q&A with RAG**; hallucinations cut ~30%, contextual accuracy >85%.
- Deployed as **Flask** API with web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring with reusable HITL escalation patterns - direct analog to lending intake + compliance validation + fraud risk scoring in financial services.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts, HITL

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time translation extension shipped to the Chrome Web Store. Per-provider abstraction across 4 AI vision providers with failure isolation. 29 sites supported across Manifest V3 service workers.

TypeScript, Manifest V3, Service Workers, Claude / GPT-4o / GPT-4.1 / Ollama

Dream Decoder - Multimodal Creative Platform

FastAPI + React + multimodal API orchestration. Intermediate prompt-transformation layer improved contextual alignment ~30% and first-pass image success ~25-30%.

FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

EDUCATION

Master's Degree - Kent State University

Aug 2023 - May 2025

Focus: Applied Machine Learning, NLP, Database Systems. STEM degree (Blue Card / sponsorship-eligible).

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

LLM / GenAI: Agentic workflows, RAG, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, HITL escalation, audit-ready reasoning

AI / ML: scikit-learn, XGBoost, PyTorch, LangChain, LlamaIndex, Hugging Face Transformers, Diffusers, YOLOv8

Backend / Infra: FastAPI, Flask, REST, Docker, Jenkins, CI/CD, Grafana, structured logging, observability

Data: PostgreSQL + pgvector, SQLite, T-SQL, Pandas, NumPy, EHR preprocessing, data lineage

Domains: Regulated-domain AI (HIPAA-conscious healthcare, oil & gas ERP), Applied AI, Agentic Workflows, Workflow Automation